

Getting Started with Fusion

Make Your Own Name Tag

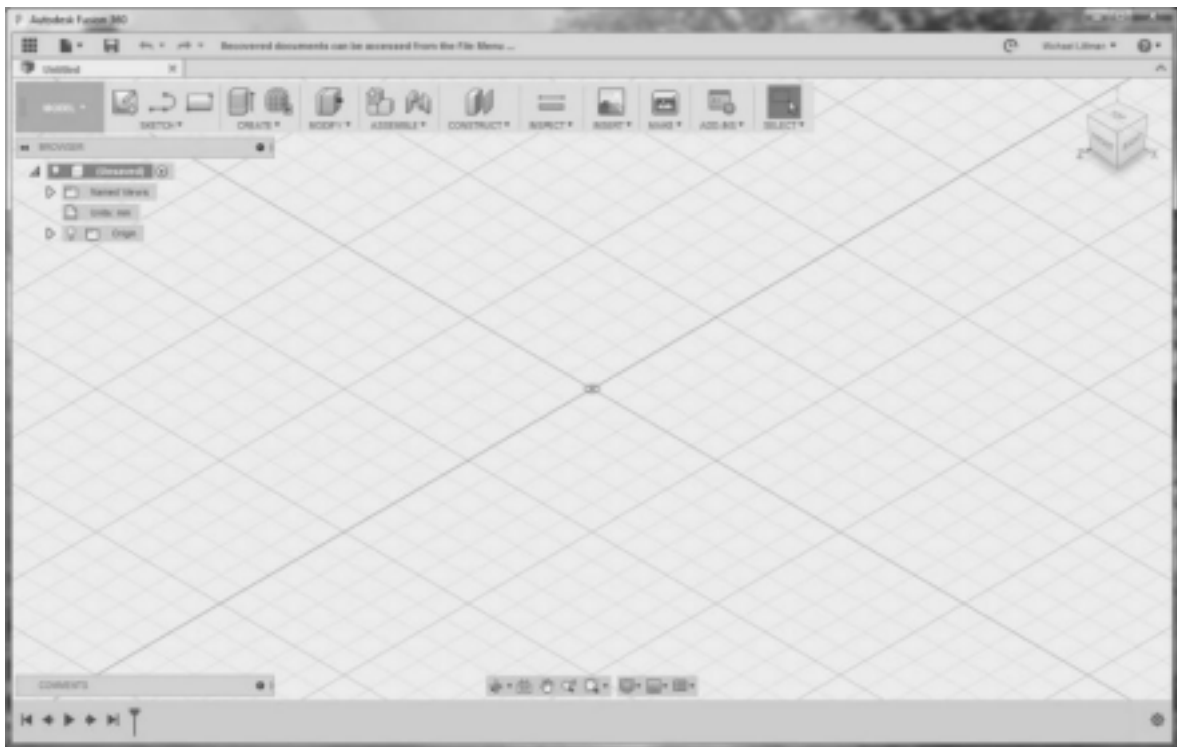
A. Introduction

Design a name tag in Fusion, then get it ready for 3D printing. You will make a base, add your name and create a hole for a keyring.

No previous experience needed. Bring a laptop with Fusion installed and sign in before the workshop.

B. Starting a New Design

1. Open Fusion and choose **New Design**.
2. Use the **Design** workspace and **Solid** tab.
3. Under **Document Settings**, set Units to **mm**.
4. Save your design as **YourName_NameTag**.

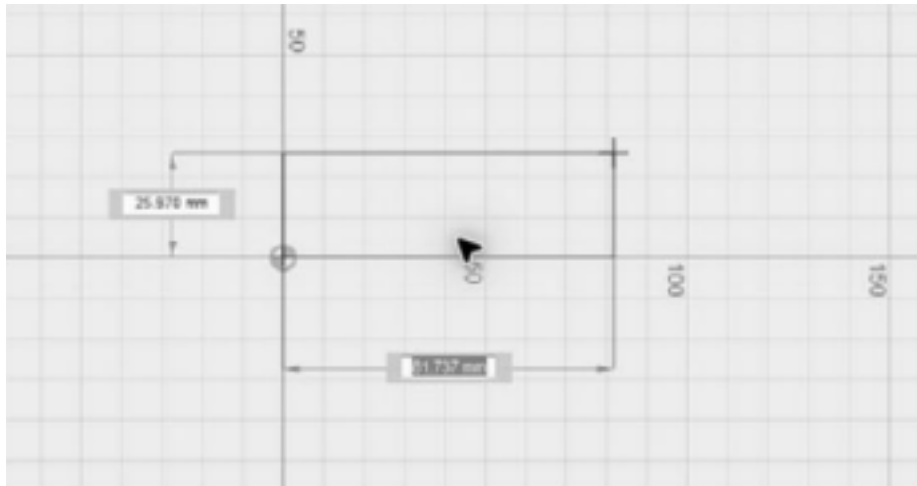


The Fusion workspace: tools at the top, your model in the middle.

Scroll to zoom. Use the **ViewCube** in the top-right corner to change your view. Screenshots show an earlier Fusion interface. Follow the tool names and dimensions in this handout.

C. Draw the Base

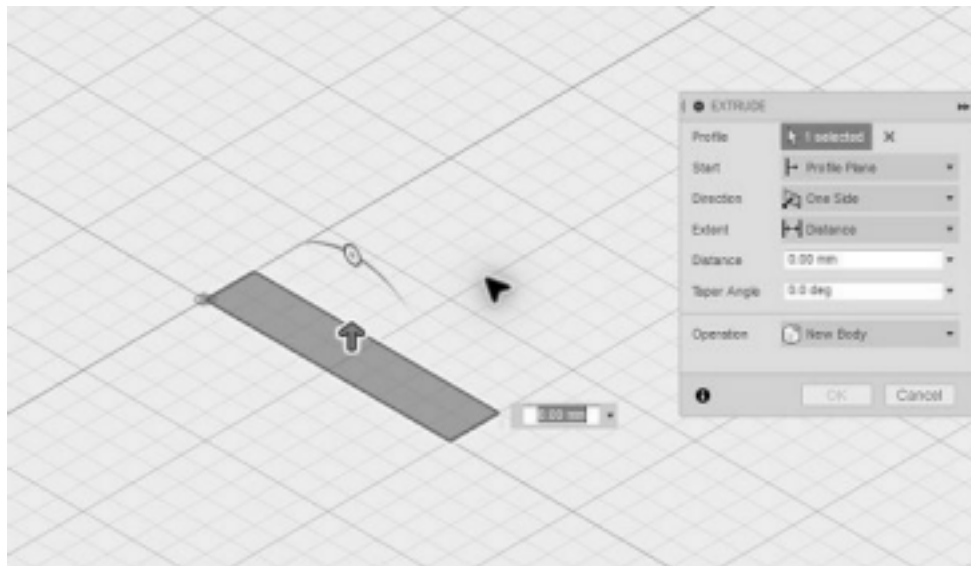
1. Click **Create Sketch** and choose the **XY plane**.
2. Choose **Rectangle > 2-Point Rectangle**.
3. Click the origin, then draw a rectangle **70 mm wide and 25 mm tall**. Use **D** to dimension the edges if needed.



Drawing a rectangle. Use 70 x 25 mm for your tag.

D. Make It 3D

1. Click **Finish Sketch**, then **Extrude** (shortcut **E**).
2. Select the filled rectangle and enter **3 mm**.
3. Set Operation to **New Body**, then click **OK**.



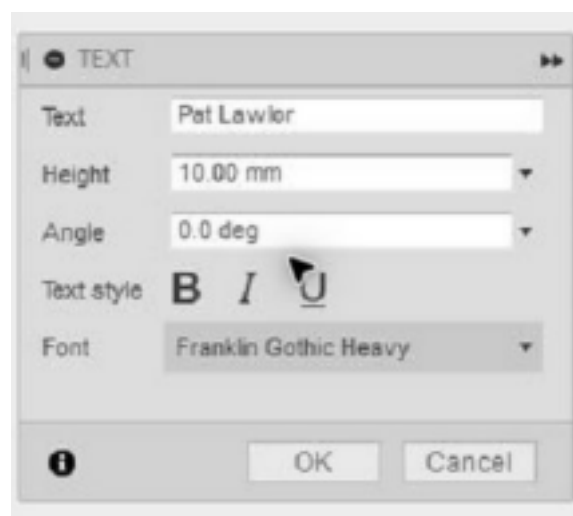
Select the rectangle, then enter the thickness in Distance.

Extrude gives a flat sketch thickness. You now have the base of your tag!

2

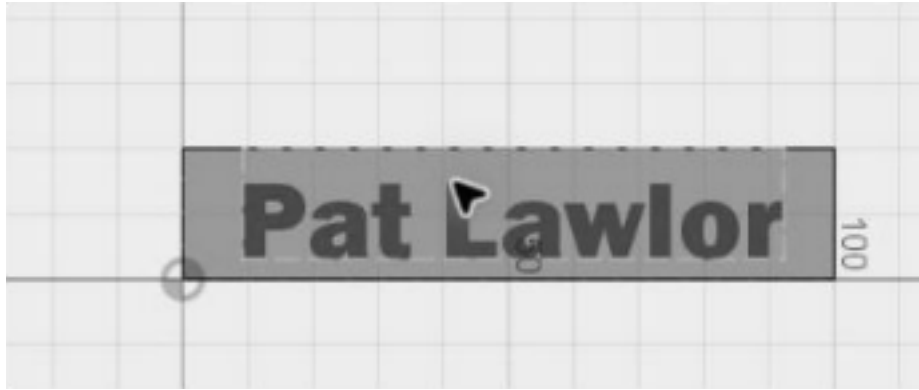
E. Add Your Name

1. Select the **top face** of the base and click **Create Sketch**.
2. Choose **Create > Text** and draw a text box.
3. Type your name. Start with a bold font and a height of **8 mm**. If the field uses expressions, enter your name in single quotes.



The Text box lets you choose your wording, size and font.

4. Move the name inside the base. Leave **2 mm around the edges** and about **12 mm on the left** for the hole. Make the text smaller if needed.



Keep your name inside the base, with space around it.

F. Raise the Letters

1. Click **OK**, then **Finish Sketch**.
2. Press **E** and select the sketch text.
3. Extrude **1 mm upwards**. Choose **Join**, then **OK**.

Join attaches the letters to the base so they print as one piece.

3

G. Add a Keyring Hole

1. Start a sketch on the **top face of the base**.
2. Choose **Center Diameter Circle**. Draw a **5 mm** circle on the left, clear of the letters. Keep its centre about **6 mm from the left edge** and halfway up the tag.
3. Finish Sketch, press **E** and select the circle.
4. Choose **Cut** and **All**. Point the extrusion down through the base, then click **OK**.

YOUR NAME

5 mm hole

Keep the letters clear of the hole

Your finished tag should look something like this.

H. Save for 3D Printing

1. Save your Fusion design.

2. Under **Bodies** on the left, right-click your tag and choose **Save As Mesh**. 3. Choose **STL (Binary)** and **Millimeter**. Save it as **YourName_NameTag.stl**.
4. Show a helper your file. They will check the size and prepare it for printing with the **flat base down and letters facing up**.

I. Try Something Different

Try a nickname, add a simple shape, or round the corners using **Fillet**. Keep your lettering bold and easy to read.

All done? Check your spelling and ask about collecting your print.

Screenshot source: Princeton University, Joseph Henry Project.
Tool reference: Autodesk Fusion Help.